

Documentation of Post Stratification Weighting

MIDUS Baseline (M1), N=3,487

This document describes how post-stratification weights were calculated for the MIDUS-Baseline (M1) National RDD sample. Two sets of weights were created for respondents who completed the phone interview (N=3,487) and the subsample who completed the SAQ (N=3,034). The M1 weights described in this document were created in order to coincide with the post-stratification weights created for the MIDUS Refresher (MR) sample. The M1 weights described here use a transparent post-stratification scheme allowing both weighted datasets to be combined, compared, and analyzed.

The sets of M1 weights described in this document were created using a population-based adjustment called a post-stratification weight. It is a *post*-stratification weight because it is computed *after* the data are collected. The *stratification* part comes from the use of various known strata (such as age group, or sex distribution) of the population to which the new data are adjusted to better conform to the population's parameters.

In order to calculate a post-stratification weight, an auxiliary dataset is needed to which the new data are compared. M1 used the October 1995 Current Population Survey (CPS) data derived from the Census Bureau. CPS data show the demographic characteristics of the U.S population at a comparable time to which the MR data were collected. (See www.census.gov/cps/).

The first step in creating the M1 post-stratification weights was to decide which demographic variables to correct for in the M1 sample. For the M1 Phone (N= 3,487) sample and the MR SAQ subsample (N= 3,034), the sample percentages among for the five demographic variables (Sex, Race, Age, Education, and Marital Status) were compared to the population percentages (Table 1). The largest discrepancies observed between the sample and population percentages were among Education and Marital Status categories. For the M1 SAQ subsample, there were also discrepancies in Sex, Age and Race. Therefore, the M1 weights discussed in this document are multivariate post-stratification weights that simultaneously correct for Sex, Race, Age, Education, and Marital Status. Two additional multivariate weights were created. One is a multivariate weight that corrects for Sex, Age, and Education only, and the other is a multivariate weight that corrects for Race, Age, and Education.

To calculate simple univariate weights, the population proportion among strata for each demographic variable was divided by the sample proportion, resulting in five univariate weights for Sex, Race, Age, Education, and Marital Status. These univariate weights are displayed in Table 1 below.

Table 1. Comparison of M1 Phone sample SAQ subsample with the Current Population Survey (October 1995) and constructed univariate weights.

MIDUS Baseline	<u>CPS (October 1995)</u> ¹	<u>Unweighted M1 Phone data</u>	<u>Unweighted M1 SAQ data</u>	<u>Univariate Weights M1 Phone data</u> ²	<u>Univariate Weights M1 SAQ data</u> ³
	(N=79,628)	(N=3,487)	(N=3,034)	(N=3,487)	(N=3,034)
	%	%	%		
SEX					
Male	47.5	49.4	48.5	0.98	0.98
Female	52.5	50.7	51.5	1.02	1.01
RACE					
White	84.1	86.5	87.7	0.97	0.96
Others	15.9	13.5	12.3	1.18	1.28
AGE					
25-34	25.4	22.5	20.5	1.20	1.24
35-44	27.4	24.6	24.3	1.11	1.13
45-54	21.0	23.4	24.2	0.88	0.87
55-64	13.8	18.9	19.8	0.71	0.70
65-74	12.4	10.8	11.2	1.11	1.08
EDUCATION					
12 years or less	49.5	40.2	39.2	1.26	1.27
13 – 15 years	25.8	31.4	31.2	0.83	0.82
16 years or more	24.8	28.5	29.6	0.83	0.83
MARITAL					
Married	67.4	62.3	64.0	1.06	1.05
Unmarried	32.6	37.7	36.0	.90	0.91

¹ CPS data filtered by age: ≥ 25 & ≤ 74

² Values are based on the univariate weights: A1PWGHT1 A1PWGHT2 A1PWGHT3 A1PWGHT4 A1PWGHT5

³ Values are based on the univariate weights: A1SWGHT1 A1SWGHT2 A1SWGHT3 A1SWGHT4 A1SWGHT5

To create the multivariate post-stratification weight that simultaneously adjusts for Sex, Race, Age, Education and Marital status, the population proportion by Sex, Race, Age, Education, and Marital Status was divided by the sample proportion. Table 2 shows the multivariate post-stratification weights for the various strata that were used for the entire M1 Phone (N=3,487) sample (A1PWGHT6).

Table 2. Post-stratification weights for M1 Phone Sample created based on various strata among Sex, Race, Age, Education, and Marital Status.

Males							
Marital Status	Race	Education	Age				
			25-34	35-44	45-54	55-64	65-74
Married	White	12 years or less	1.6	1.2	1.0	0.9	1.5
		13-15 years	1.0	0.9	0.9	0.7	0.7
		16 years or more	0.9	1.0	0.7	0.6	0.6
	Other	12 years or less	1.4	1.3	1.5	1.3	1.6
		13-15 years	0.9	1.1	0.7	0.9	1.8
		16 years or more	0.8	1.3	1.3	0.7	1.6
Not Married	White	12 years or less	2.5	1.4	1.2	0.8	1.1
		13-15 years	1.0	1.0	0.5	0.5	0.7
		16 years or more	0.8	0.6	0.5	0.4	0.4
	Other	12 years or less	2.4	2.7	1.2	2.1	2.5
		13-15 years	0.7	0.7	0.5	1.1	0.5
		16 years or more	0.9	0.8	0.4	0.5	0.5
Females							
Marital Status	Race	Education					
Married	White	12 years or less	1.6	1.6	1.1	0.9	1.7
		13-15 years	1.0	1.2	1.0	0.5	1.0
		16 years or more	1.4	1.1	0.9	0.6	1.1
	Other	12 years or less	1.1	1.6	1.7	1.1	4.1
		13-15 years	1.1	1.0	2.0	2.9	0.9
		16 years or more	1.4	1.2	2.3	0.91	1.0
Not Married	White	12 years or less	1.2	1.0	1.0	0.7	1.3
		13-15 years	1.0	0.7	0.5	0.3	0.5
		16 years or more	0.9	1.0	0.5	0.4	1.1
	Other	12 years or less	1.6	1.3	1.9	1.0	1.8
		13-15 years	0.7	1.3	1.3	0.4	0.6
		16 years or more	1.3	0.6	0.5	0.3	1.2

The multivariate post-stratification weight that is described was applied to each of the demographic variables to determine how well the multivariate weight adjusted for discrepancies in sex, race, age, education, and marital status. Table 3 shows the unweighted and weighted strata distributions for the M1 Phone (N=3,487) sample and the M1 SAQ subsample (N=3,034).

Table 3. Comparison of un-weighted and multivariate post-stratification weighted M1 Phone sample and M1 SAQ subsample with the Current Population Survey (October 1995).

	<u>CPS</u> <u>(October</u> <u>1995)</u>	<u>Unweighted</u> <u>M1 Phone</u> <u>data</u>	<u>Weighted</u> <u>M1 Phone</u> <u>data</u> ²	<u>Unweighted</u> <u>M1 SAQ</u> <u>data</u>	<u>Weighted</u> <u>M1 SAQ</u> <u>data</u> ³
MIDUS Baseline	(N=79,628)	(N=3,487)	(N=3,487)	(N=3,034)	(N=3,034)
	%	%	%	%	%
SEX					
Male	47.5	49.4	48.4	48.5	47.7
Female	52.5	50.7	51.6	51.5	52.3
RACE					
White	84.1	86.5	84.1	87.6	84.1
Others	15.9	13.5	15.9	12.4	15.9
AGE					
25-34	25.4	22.5	26.3	20.5	25.3
35-44	27.4	24.6	27.0	24.3	27.4
45-54	21.0	23.4	20.9	24.2	21.1
55-64	13.8	18.9	13.9	19.8	14.1
65-74	12.4	10.8	11.9	11.2	12.1
EDUCATION					
12 years or less	49.5	40.2	49.5	39.2	49.6
13 – 15 years	25.8	31.4	26.6	31.2	25.8
16 years or more	24.8	28.5	23.9	29.6	24.6
MARITAL					
Married	67.4	62.3	65.6	64.0	67.2
Unmarried	32.6	37.7	34.4	36.0	32.8

² Values are based on multivariate weight labeled: A1PWGHT6

³ Values are based on multivariate weight labeled: A1SWGHT6